



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 32

Level 2 — Intermediate · Measurement I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Measurement I — km/m, Multi-Unit Conversions
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	32 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 32 deepens measurement with kilometre-metre conversions, the mm, tricky zero cases (6 kg 50 g = 6,050 g — NOT 6,500), borrowing across units (5 m – 2 m 35 cm), unit-price scaling (250 g for ₹200 → 1 kg?), and ordering mixed-unit weights. These zero-trap conversions are among the most missed questions in real IMO papers.

Learning Objectives

- Convert $\text{km} \leftrightarrow \text{m}$, $\text{kg} \leftrightarrow \text{g}$, $\text{L} \leftrightarrow \text{mL}$ and $\text{cm} \leftrightarrow \text{mm}$, including zero-trap cases.
- Add and subtract mixed-unit measurements with borrowing.
- Multiply and divide measurements in laps, packs, mugs and doses.
- Scale unit prices from grams to kilograms.
- Order mixed-unit quantities by converting to one unit.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. Convert: **3 km 250 m = ___ m**

- (A) 3,250 (B) 3,025
(C) 32,500 (D) 325

Q2. Convert: **4,725 m = ___ km ___ m**

- (A) 4 km 725 m (B) 47 km 25 m
(C) 4 km 72 m (D) 4 km 7 m

Q3. Convert: **6 kg 50 g = ___ g**

- (A) 650 (B) 6,050
(C) 6,500 (D) 6,005

Q4. Convert: **2 L 75 mL = ___ mL**

- (A) 275 (B) 2,750
(C) 2,075 (D) 20,075

Q5. Which is the LONGEST distance?

- (A) 2 km (B) 2,500 m
(C) 25,000 cm (D) 2 km 300 m

Q6. What is **half of 5 kg**, in grams?

- (A) 250 g (B) 2,500 g
(C) 500 g (D) 25,000 g

Q7. A runner completes 4 rounds of a 400 m track. How far does he run in km and m?

- (A) 1 km 200 m (B) 1 km 600 m
(C) 1 km 400 m (D) 2 km

Q8. Subtract: **5 m – 2 m 35 cm**

- (A) 2 m 65 cm (B) 3 m 65 cm
(C) 2 m 35 cm (D) 3 m 35 cm

Q9. A jar contains 1 L of juice. Glasses of 200 mL are filled from it. How many FULL glasses can be poured?

- (A) 4 (B) 5
(C) 6 (D) 10

Q10. Add: **2 kg 750 g + 1 kg 500 g**

- (A) 3 kg 250 g (B) 4 kg 250 g
(C) 4 kg 50 g (D) 3 kg 750 g



Q11. A bag of 10 kg rice is repacked into smaller packs of 2 kg 500 g each. How many packs are made?

- (A) 3 (B) 4
(C) 5 (D) 8

Q12. Which is HEAVIER, and by how much: a 3 kg 200 g parcel or a 2 kg 850 g parcel?

- (A) First, by 350 g (B) Second, by 350 g
(C) First, by 450 g (D) First, by 650 g

Q13. A bottle of 1 L 500 mL water is shared equally between 2 friends. Each gets:

- (A) 500 mL (B) 650 mL
(C) 750 mL (D) 1 L

Q14. The distance from Riya's home to school is 1 km 800 m. She goes by cycle and returns on foot daily. How far does she travel each day?

- (A) 1 km 800 m (B) 2 km 600 m
(C) 3 km 600 m (D) 3 km 800 m

Q15. 1 cm = 10 mm. How many millimetres are there in 4 cm 5 mm?

- (A) 45 mm (B) 405 mm
(C) 450 mm (D) 9 mm

Q16. A shopkeeper sells 250 g of cashews for ₹200. What is the price of 1 kg of cashews?

- (A) ₹400 (B) ₹600
(C) ₹800 (D) ₹1,000

Q17. A tank of 20 L is filled using a 500 mL mug. How many mugs are needed?

- (A) 20 (B) 30
(C) 40 (D) 50

Q18. Arrange from LIGHTEST to HEAVIEST: 1,050 g · 1 kg 5 g · 1 kg 500 g

- (A) 1 kg 5 g · 1,050 g · 1 kg 500 g (B) 1,050 g · 1 kg 5 g · 1 kg 500 g
(C) 1 kg 500 g · 1,050 g · 1 kg 5 g (D) 1 kg 5 g · 1 kg 500 g · 1,050 g

Q19. A doctor prescribes 5 mL of syrup three times a day. How much syrup is taken in 4 DAYS?

- (A) 15 mL (B) 20 mL
(C) 60 mL (D) 100 mL

Q20. A wall is 2 m 40 cm high. A table 80 cm high stands against it. How much of the wall is ABOVE the table?

- (A) 1 m 60 cm (B) 1 m 40 cm
(C) 2 m 60 cm (D) 1 m 80 cm



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	A	B	C	B	B	B	A	B	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	A	C	C	A	C	C	A	C	A

Step-by-Step Solutions

Q1. Correct answer: (A) $3 \text{ km} = 3,000 \text{ m}; + 250 \text{ m} = 3,250 \text{ m}.$ **Q2. Correct answer: (A)** $4,725 = 4,000 + 725 \rightarrow 4 \text{ km } 725 \text{ m}.$ **Q3. Correct answer: (B)** $6 \text{ kg} = 6,000 \text{ g}; + 50 \text{ g} = 6,050 \text{ g}.$ (Careful — NOT 6,500!)**Q4. Correct answer: (C)** $2 \text{ L} = 2,000 \text{ mL}; + 75 = 2,075 \text{ mL}.$ **Q5. Correct answer: (B)**Convert to metres: $2,000 \text{ m}; 2,500 \text{ m}; 250 \text{ m}; 2,300 \text{ m}.$ The longest is $2,500 \text{ m}.$ **Q6. Correct answer: (B)** $5 \text{ kg} = 5,000 \text{ g}.$ Half = $2,500 \text{ g} (= 2 \text{ kg } 500 \text{ g}).$ **Q7. Correct answer: (B)** $4 \times 400 = 1,600 \text{ m} = 1 \text{ km } 600 \text{ m}.$ **Q8. Correct answer: (A)** $500 \text{ cm} - 235 \text{ cm} = 265 \text{ cm} = 2 \text{ m } 65 \text{ cm}.$ **Q9. Correct answer: (B)** $1,000 \div 200 = 5 \text{ glasses}.$ **Q10. Correct answer: (B)** $2,750 + 1,500 = 4,250 \text{ g} = 4 \text{ kg } 250 \text{ g}.$ **Q11. Correct answer: (B)** $10,000 \div 2,500 = 4 \text{ packs}.$ **Q12. Correct answer: (A)** $3,200 - 2,850 = 350.$ The first parcel is heavier by $350 \text{ g}.$ **Q13. Correct answer: (C)** $1,500 \div 2 = 750 \text{ mL each}.$ **Q14. Correct answer: (C)** $2 \times 1,800 \text{ m} = 3,600 \text{ m} = 3 \text{ km } 600 \text{ m}.$ **Q15. Correct answer: (A)** $4 \text{ cm} = 40 \text{ mm}; + 5 \text{ mm} = 45 \text{ mm}.$



Q16. Correct answer: (C)

1 kg = 4 × 250 g, so price = 4 × ₹200 = **₹800**.

Q17. Correct answer: (C)

20 L = 20,000 mL. $20,000 \div 500 =$ **40 mugs**.

Q18. Correct answer: (A)

Convert: 1,050 g; 1,005 g; 1,500 g. Order: **1,005 < 1,050 < 1,500**, i.e., 1 kg 5 g · 1,050 g · 1 kg 500 g.

Q19. Correct answer: (C)

Per day = $5 \times 3 = 15$ mL. In 4 days = $15 \times 4 =$ **60 mL**.

Q20. Correct answer: (A)

$240 - 80 = 160$ cm = **1 m 60 cm**.

